

$\sqrt{bc}$ inversion FTW!

AHARSHI ROY

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“Were you rushing or were you dragging?”

§1 Introduction

Today, we are going to learn about a special type of inversion, typically used in triangle configuration problems, called $\sqrt{bc}$ inversion (obviously it will be beneficial if you had some background knowledge about inversion, you can learn it from say EGMO Chapter 8, which is free from Evan Chen’s website).

This is a bit different than your normal inversion maps, because it is actually composed of two transformations! Without further ado, we are going to give the statement.

Lemma ($\sqrt{bc}$ inversion)

If we say to $\sqrt{bc}$ invert at X in $\triangle XYZ$, then we are composing these two maps

- Invert around the circle with center X and radius $\sqrt{XY \cdot XZ}$.
- Perform a reflection around the internal angle bisector of $\angle YXZ$.

Remark. If you have already seen the statement of $\sqrt{bc}$ inversion before, then you might be a bit confused why I wrote $\triangle XYZ$ instead of usual $\triangle ABC$. This is because sometimes you want “ $\sqrt{bc}$ invert” per say. b and c refer to the side lengths AC and AB respectively but sometimes we will “ $\sqrt{bc}$ invert” at C or some other vertex! To save me the trouble of writing $\sqrt{ab}$ invert or even worse, write “Now we will invert around circle with center C and radius $\sqrt{CA \cdot CB}$ and reflect around the internal angle bisector of $\angle ACB$ ” all the time; to remove ambiguity also, I write like this.

§2 Theory

§2.1 All the points we are going to use

Ok this is the long list of points we are going to use. By the way, some of these bullet points should end with “.... of $\triangle ABC$ ”.

- I, G, O, H is the incenter, centroid, circumcenter and orthocenter respectively.
- $\triangle I_A I_B I_C$ is the ex central triangle.
- $\triangle DEF$ is the orthic triangle.

- A' is the A -antipode.
- K is the point on (ABC) such that $(A, K; B, C) = -1$.
- $\triangle MNP$ is the medial triangle.
- T_A is the A -mixtilinear intouch point.
- L_A is the major arc midpoint of $\widehat{BC}$.
- U, V, W is the tangency points of the A -excircle with $\overline{BC}, \overline{AB}, \overline{AC}$ respectively.
- A_2 is the point on (ABC) such that $\overline{AA_2} \parallel \overline{BC}$.

§2.2 Where do the important points go to?

Now to honor that boring long remark, we will say *throughout this section, we are going to $\sqrt{bc}$ invert at A in $\triangle ABC$.*

Claim — B swaps with C . Hence $\overline{BC}$ swaps with (ABC) .

Subproof. This is easy, as B^* (means image of B under this map) must lie on the isogonal line of $\overline{AB}$ which is $\overline{AC}$ and $AB \cdot AB^* = AB \cdot AC$ and hence the result follows. $\square$

Claim — I swaps with I_A .

Subproof. We can do this by a length bash, but I would like to do it by an angle argument to show how it is useful sometimes.

We know that $\angle ABI = \frac{\angle B}{2}$ hence $\angle AI^*C = \frac{\angle B}{2}$. But an easy angle chase gives us

$$\angle AI_A C = 180^\circ - \angle I_A A C - \angle A C I_A = 180^\circ - \frac{\angle A}{2} - 90^\circ - \frac{\angle C}{2} = \frac{\angle B}{2}$$

And see that $\overline{AI^*}$ is the isogonal line of $\overline{AI}$ which is itself and we are done. $\square$

Claim — I_B and I_C swap with each other.

Subproof. Sometimes it is view a triangle config problem from the eye of the excentral triangle (general advice) as we can then exploit Orthocenter-Incenter duality which might be useful. Anyways, this is a good example of that.

See that I is the orthocenter of $\triangle I_A I_B I_C$; if we reflect I over A , it lies on $(I_A I_B I_C)$ (see EGMO Chapter 1, if you didn't know that) and hence if we denote that point by A_1 , we get by PoP that

$$AI_B \cdot AI_C = AI_A \cdot AA_1 = AI_A \cdot AI = AB \cdot AC$$

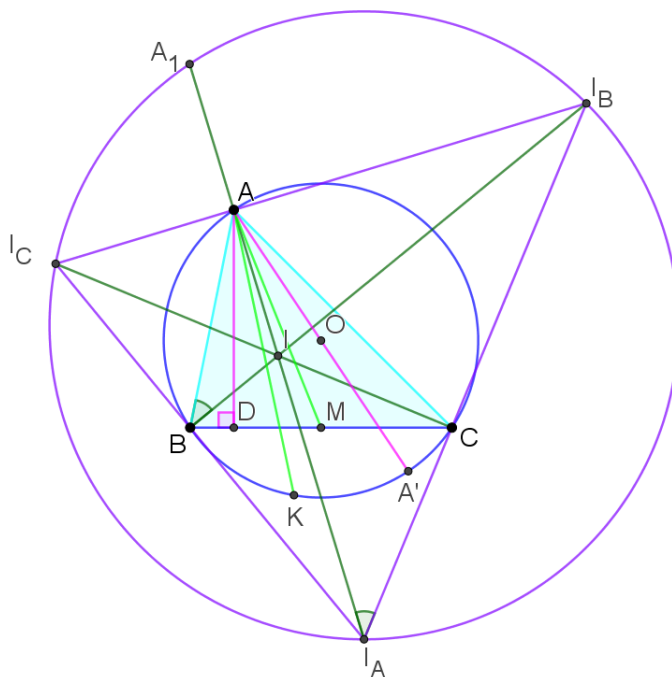
And done. $\square$

Sometimes isogonal line arguments are enough to get some crucial claims. The next two pairings are examples of that.

Claim — A' swaps with D .

Remark. From this we can also get where O goes, as O is the midpoint of $\overline{AA'}$; O^* must be the reflection of A over D . But this might be more useful in something called $\sqrt{\frac{bc}{2}}$ inversion, we will have a short version of it later.

Claim — K and M swap with each other.



Now we will dedicate this minisection, just on Mixtilinear circles, especially the A -mixtilinear incircle (call it Ω).

Claim — We have that

- The A -mixtilinear incircle swaps with the A -excircle.
- The A -mixtilinear excircle swaps with the incircle.
- A and I map to each other under inversion about Ω .
- L_A, I, T_A are collinear.

Subproof. The first one is just because the fact that inversion preserves tangencies.

See that Ω is tangent to $\overline{AB}$, $\overline{AC}$ and (ABC) and so Ω^* is tangent to $\overline{AC}$, $\overline{AB}$ and $\overline{BC}$; and see that Ω is “inside” $\triangle ABC$ and so Ω^* is “outside” $\triangle ABC$, and so it is just the A -excircle. The next bullet point is the same argument.

Now for the third point, see that we just need to prove that $(AVWI_A)$ is a circle and $(A, I_A; V, W) = 1$. The first thing is obvious due to right angles and see that $\overline{AI_A}$

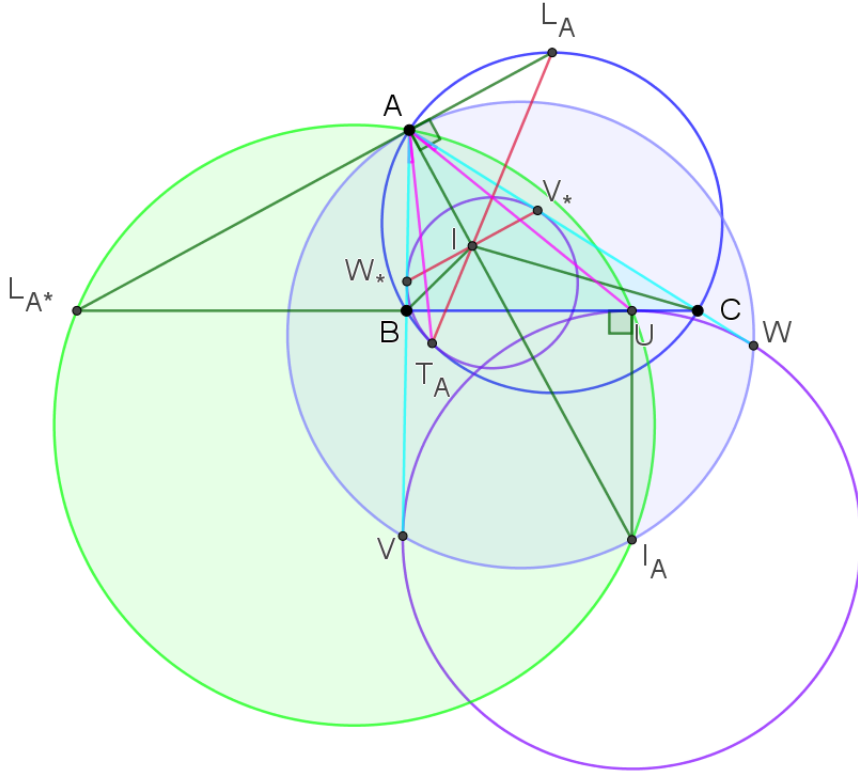
bisects $\angle VAW$ and so it is a kite! Meaning we have that

$$-1 = (A, I_A; V, W) = (A^*, I_A^*, V^*, W^*) = (\infty, I; V^*, W^*)$$

and hence I is the midpoint of the touch chord as required.

For the last bullet point, first see that the first bullet point tells us that T_A swaps with U (also means that $\overline{AT_A}$ and $\overline{AU}$ are isogonal) and see that L_A swaps with the point on $\overline{BC}$ such that $\angle L_A^* AI = 90^\circ$. So we need to prove that $(L_A^* AI_A U)$ is cyclic which is obvious due to right angles. $\square$

Remark. See that if we define T_B and T_C similarly as T_A , then $\overline{AT_A} \cap \overline{BT_B} \cap \overline{CT_C}$ exists because it is the isogonal conjugate of the Nagel Point!



Claim — The polar of $\overline{AK}$ with respect to (ABC) swaps with A_2 .

Subproof. Try doing this one on your own (look at tangencies). $\square$

Now you might be thinking “Why isn’t this guy talking about the two most important triangle centers?” The reason is that they do not go to “very good points”.

But we still do it, because it seems a bit incomplete without them.

Claim — G^* is the point on line $\overline{AK}$ such that $AK = 2AG^*$.

Subproof. See that $(A, M; G, \infty) = -\frac{1}{2}$ and so $(\infty, K; G, A) = -\frac{1}{2}$ hence the result. So see that one can actually see this as the reflection of the A -Dumpty point over K . $\square$

§2.3 Short section on $\sqrt{\frac{bc}{2}}$ inversion

This is exactly how it sounds.

Lemma ($\sqrt{\frac{bc}{2}}$ inversion)

If we say to $\sqrt{\frac{bc}{2}}$ invert at X in $\triangle XYZ$, it is composed of two maps again

- Invert around the circle with center X and radius $\sqrt{\frac{XY \cdot XZ}{2}}$.
- Perform a reflection around the angle bisector of $\angle YXZ$.

This is basically just one more map that we do after the $\sqrt{bc}$ inversion, that is taking a $\frac{1}{2}$ homothety at X .

Why am I saying this? Because sometimes this is more useful! I am just quickly going to state where each of the points go.

And oh before that! *Throughout this section, we are going to $\sqrt{\frac{bc}{2}}$ invert at A in $\triangle ABC$.*

Claim — We have that

- B swaps with N , C swaps with P .
- $\overline{NP}$ swaps with (ABC) and $\overline{BC}$ swaps with (APN) .
- O swaps with D .

Subproof. All are trivial from the last section. □

One of the main reasons, we use this “weirder” force-lay inversion rather than the former is because of this.

Claim — The nine point circle of $\triangle ABC$ swaps with (BOC) .

Subproof. See that (NDP) swaps with (BOC) . □

§3 Problems

Just for fun, I have thrown in a “non $\sqrt{bc}$ problem”, a problem which I think cannot be solved by $\sqrt{bc}$ inversion elegantly, but it has one of the nicest solutions to a problem ever in my opinion. This is so that, you guys do not think $\sqrt{bc}$ is a tool which nukes a problem and apply it all the triangle geo problems you see!

Problem 3.1 (4-G-UnS-0). If D_A is the A -Dumpty point where it is defined as $(AO) \cap (BOC)$. Then prove that D_A is the midpoint of $\overline{AK}$.

Problem 3.2 (4-G-UnS-1 (MOP 2017)). Let $\triangle ABC$ be a triangle with circumcircle γ and denote by M and N the midpoints of $\overline{AB}$ and $\overline{AC}$. Suppose the circumcircles of triangles ABN and ACM meet again at P . Prove that $\overline{AP}$ is concurrent with the pole of $\overline{BC}$ with respect to γ .

Problem 3.3 (Serbian National Olympiad 2013/3). Let M , N and P be midpoints of sides BC , AC and AB , respectively, and let O be circumcenter of acute-angled triangle ABC . Circumcircles of triangles BOC and MNP intersect at two different points X and Y inside of triangle ABC . Prove that $\angle BAX = \angle CAY$.

Problem 3.4 (All Russian Maths Olympiad 2009/10.2). Let $\triangle ABC$ have its internal angle bisector of $\angle A$ cut $\overline{BC}$ at D and (ABC) at N . Now let circle with diameter $\overline{DE}$ cut (ABC) again at K . Prove that $\overline{AK}$ is the A -symmedian of $\triangle ABC$.

Problem 3.5 (USA TST 2016/2). Let ABC be a scalene triangle with circumcircle Ω , and suppose the incircle of ABC touches BC at D . The angle bisector of $\angle A$ meets BC and Ω at E and F . The circumcircle of $\triangle DEF$ intersects the A -excicle at S_1 , S_2 , and Ω at $T \neq F$. Prove that line AT passes through either S_1 or S_2 .

Problem 3.6 (ELMO 2010/6). Let ABC be a triangle with circumcircle ω , incenter I , and A -excenter I_A . Let the incircle and the A -excicle hit BC at D and E , respectively, and let M be the midpoint of arc BC without A . Consider the circle tangent to BC at D and arc BAC at T . If TI intersects ω again at S , prove that SI_A and ME meet on ω .

Problem 3.7 (ELMO 2014/5). Let ABC be a triangle with circumcenter O and orthocenter H . Let ω_1 and ω_2 denote the circumcircles of triangles BOC and BHC , respectively. Suppose the circle with diameter $\overline{AO}$ intersects ω_1 again at M , and line AM intersects ω_1 again at X . Similarly, suppose the circle with diameter $\overline{AH}$ intersects ω_2 again at N , and line AN intersects ω_2 again at Y . Prove that lines MN and XY are parallel.

Problem 3.8 (ELMO 2016/6). My dog Tiger is now learning olympiad geometry, but he is a newbie so help him please. In a triangle ABC with $AB \neq AC$, let its incircle be tangent to sides BC , CA , and AB at D , E , and F , respectively. The internal angle bisector of $\angle BAC$ intersects lines DE and DF at X and Y , respectively. Let S and T be distinct points on side BC such that $\angle XSY = \angle XTY = 90^\circ$. Finally, let γ be the circumcircle of $\triangle AST$.

(a) Help Tiger show that γ is tangent to the circumcircle of $\triangle ABC$.

(b) Help Tiger show that γ is also tangent to the incircle of $\triangle ABC$.

Problem 3.9 (Japan TST 2022/6). Given is a circle Γ with diameter MN and a point A inside Γ . The circle with center N , passing through A , meets Γ at B and C . Let

$P, Q \in BC$, such that $\angle BAP = \angle QAC$. The lines NP, NQ meet Γ at X, Y , respectively. Prove that AM, PY, QX are concurrent.

Problem 3.10 (Taiwan TST Round 3 2014/3). Let ABC be a triangle with circumcircle Γ and let M be an arbitrary point on Γ . Suppose the tangents from M to the incircle of ABC intersect $\overline{BC}$ at two distinct points X_1 and X_2 . Prove that the circumcircle of triangle MX_1X_2 passes through the tangency point of the A -mixtilinear incircle with Γ .

Problem 3.11 (IGO 2023/A1). We are given an acute triangle ABC . The angle bisector of $\angle BAC$ cuts BC at P . Points D and E lie on segments AB and AC , respectively, so that $BC \parallel DE$. Points K and L lie on segments PD and PE , respectively, so that points A, D, E, K, L are concyclic. Prove that points B, C, K, L are also concyclic.

Problem 3.12 (4-G-UnS-2). Let $\triangle ABC$ be a triangle and define Ω_a to be the Apollonian circle passing through A and foci B and C . Define Ω_b and Ω_c similarly. Prove that $\Omega_a, \Omega_b, \Omega_c$ intersect at 2 points. The “inside” intersection is the *First Isodynamic Point* or X_{15} . So further prove that X_{15} is the isogonal conjugate to the *First Fermat Point* or X_{13} .

Problem 3.13 (4-G-UnS-3). Let there be $\triangle ABC$ with circumcenter O and incenter I . Let ω be the circle externally tangent to (BOC) and tangent to $\overline{AB}$ and $\overline{AC}$ at say M and N respectively. Prove that $\overline{MN}$ bisects $\overline{AI}$.

§4 Solutions

§4.1 4-G-UnS-0

Problem statement

If D_A is the A -Dumpty point where it is defined as $(AO) \cap (BOC)$. Then prove that D_A is the midpoint of $\overline{AK}$.

Proof. See that after $\sqrt{\frac{bc}{2}}$ inversion at A in $\triangle ABC$, we get that $D_A^* = \overline{BC} \cap (MNP) = M$. Invert back and we are done. ■

§4.2 4-G-UnS-1 (MOP 2017)**Problem statement**

Let $\triangle ABC$ be a triangle with circumcircle γ and denote by M and N the midpoints of $\overline{AB}$ and $\overline{AC}$. Suppose the circumcircles of triangles ABN and ACM meet again at P . Prove that $\overline{AP}$ is concurrent with the pole of $\overline{BC}$ with respect to γ .

Proof. Let X be intersection of (ABN) and (ACM) . We will apply $\sqrt{\frac{bc}{2}}$ inversion at A in $\triangle ABC$ to the whole problem and so we get

- M and C are swapped.
- N and B are swapped.
- (ABN) swaps with $\overline{BN}$ and similarly (ACM) swaps with $\overline{CM}$.
- $X^* = \overline{BN} \cap \overline{CM} = G$ (which is the centroid of $\triangle ABC$).

This basically means inverting back we get that since $\overline{AX}$ and $\overline{AX^*}$ is isogonal, $\overline{AX}$ is exactly the A -symmedian of $\triangle ABC$, and we are done. ■

§4.3 Serbian National Olympiad 2013/3**Problem statement**

Let M , N and P be midpoints of sides BC , AC and AB , respectively, and let O be circumcenter of acute-angled triangle ABC . Circumcircles of triangles BOC and MNP intersect at two different points X and Y inside of triangle ABC . Prove that $\angle BAX = \angle CAY$.

Proof. We apply $\sqrt{\frac{bc}{2}}$ inversion at A in $\triangle ABC$ to the whole problem and we get that

- B swaps with N and similarly C swaps with P .
- O swaps with D (which is the foot of the altitude from A to $\overline{BC}$).
- And so (BOC) swaps with (PND) which is just (MNP) .

And hence this basically means X and Y are symmetric to each other with respect to this inversion which means $\overline{AX}$ and $\overline{AY}$ are isogonal lines, and we win. ■

Remark. Overkill? Yes. I care? No.

§4.4 All Russian Maths Olympiad 2009/10.2**Problem statement**

Let $\triangle ABC$ have its internal angle bisector of $\angle A$ cut $\overline{BC}$ at D and (ABC) at N . Now let circle with diameter $\overline{DE}$ cut (ABC) again at K . Prove that $\overline{AK}$ is the A -symmedian of $\triangle ABC$.

Proof. We will apply $\sqrt{bc}$ inversion at A in $\triangle ABC$ to the problem. See that D swaps with N and since center of $\overline{DN}$ is obviously on the angle bisector of $\angle A$, we get that the circle with diameter $\overline{DN}$ stays put.

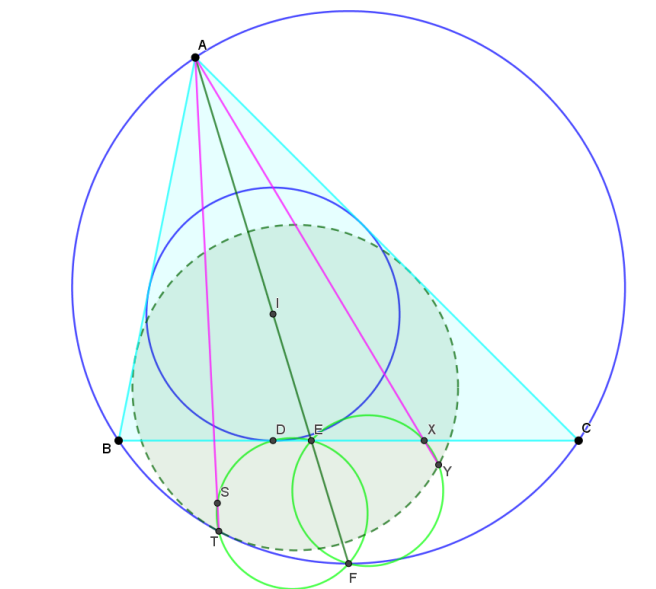
And so K swaps with $(DNK) \cap \overline{BC}$ which is the midpoint of $\overline{BC}$, call it M . And so $\overline{AM}$ and $\overline{AK}$ are isogonal and we win. ■

§4.5 USA TST 2016/2 (proposed by Evan Chen)**Problem statement**

Let ABC be a scalene triangle with circumcircle Ω , and suppose the incircle of ABC touches BC at D . The angle bisector of $\angle A$ meets BC and Ω at E and F . The circumcircle of $\triangle DEF$ intersects the A -excircle at S_1, S_2 , and Ω at $T \neq F$. Prove that line AT passes through either S_1 or S_2 .

Proof. We define some new points

- $S = \overline{AT} \cap (DEF)$.
- X is where the A -Nagel Cevian intersects $\overline{BC}$.
- $Y = \overline{AX} \cap (XEF)$.



We apply $\sqrt{bc}$ inversion at A in $\triangle ABC$ to the entire problem (and remember that it is well known that T is actually the A -Mixtilinear Touch point) and hence

- E and F are swapped.
- T and X are swapped.
- S and Y are swapped.

Now see that $\overline{SX} \parallel \overline{TY}$ since

$$AX \cdot AT = AY \cdot AS \iff \frac{AX}{AY} = \frac{AS}{AT}$$

and also $(SXTY)$ is cyclic (by power of a point) and so $\overline{AIEF} \perp \overline{TY}$ and since the center of the A -Mixtilinear incircle is on line $\overline{AI}$ we have that Y lies on that circle as well, which means its image S lies on the A -excircle, as desired. ■

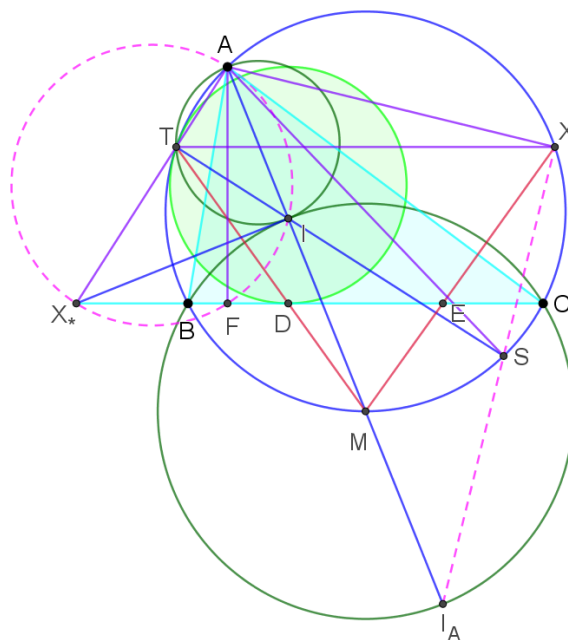
§4.6 ELMO 2010/6

Problem statement

Let ABC be a triangle with circumcircle ω , incenter I , and A -excenter I_A . Let the incircle and the A -excircle hit BC at D and E , respectively, and let M be the midpoint of arc BC without A . Consider the circle tangent to BC at D and arc BAC at T . If TI intersects ω again at S , prove that SI_A and ME meet on ω .

Proof. First we define some new points).

- $X = \overline{ME} \cap (ABC)$.
- $Y = \overline{AT} \cap \overline{BC}$.
- O is the center of (ABC) .
- R is the foot of altitude from A to $\overline{BC}$.



First we note some preliminary stuff.

- By the Shooting star lemma, we get that M, D, T are collinear, which means that T is the infamous A -Sharky-Devil point.
- This also means that S is the A -antipode.

And so see that X is the reflection of T over $\overline{MO}$. This shows us that $\overline{AT}$ and $\overline{AX}$ are isogonal.

Now comes up the meat of the problem. We apply $\sqrt{bc}$ inversion at A in $\triangle ABC$ to the problem (yay). See that

- X and Y are swapped.

- I_A and I are swapped.
- S and R are swapped.

And hence we have to prove that A, I, R, Y are concyclic. And this is equivalent to prove that $90^\circ = \angle ARY = \angle AIY$. But see that (BIC) and (AIT) are tangent to each other (since I, M and midpoint of $\overline{AI}$ are collinear obviously) and hence since

$$YT \cdot YA = YB \cdot YC \iff Y \text{ lies on the radical axis of the two circles}$$

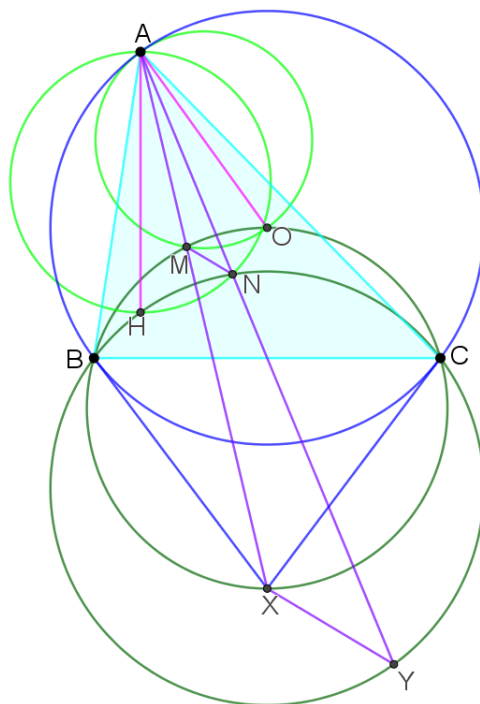
we are done. ■

§4.7 ELMO 2014/5

Problem statement

Let ABC be a triangle with circumcenter O and orthocenter H . Let ω_1 and ω_2 denote the circumcircles of triangles BOC and BHC , respectively. Suppose the circle with diameter $\overline{AO}$ intersects ω_1 again at M , and line AM intersects ω_1 again at X . Similarly, suppose the circle with diameter $\overline{AH}$ intersects ω_2 again at N , and line AN intersects ω_2 again at Y . Prove that lines MN and XY are parallel.

Proof. Surprisingly, no new points added!



We will apply $\sqrt{bc}$ inversion at A in $\triangle ABC$ to the problem and see it is well known that (BHC) and (BOC) are swapped; and that M and N are isogonal conjugates. and And hence we get that

- M and Y are swapped.
- N and X are swapped.

And so we get that

$$AM \cdot AY = AN \cdot AX \iff \frac{AM}{AN} = \frac{AX}{AY}$$

and so we are done. ■

§4.8 ELMO 2016/6**Problem statement**

My dog Tiger is now learning olympiad geometry, but he is a newbie so help him please. In a triangle ABC with $AB \neq AC$, let its incircle be tangent to sides BC , CA , and AB at D , E , and F , respectively. The internal angle bisector of $\angle BAC$ intersects lines DE and DF at X and Y , respectively. Let S and T be distinct points on side BC such that $\angle XSY = \angle XTY = 90^\circ$. Finally, let γ be the circumcircle of $\triangle AST$.

- (a) Help Tiger show that γ is tangent to the circumcircle of $\triangle ABC$.
- (b) Help Tiger show that γ is also tangent to the incircle of $\triangle ABC$.

We define some new points.

- Let G, H be the intersection points of $\overline{AI}$ and (DEF) (order does not matter).
- Let $K = \overline{AD} \cap (DEF)$.
- Let $L = \overline{BC} \cap \overline{AI}$.

Claim (Main Claim) — X and Y are exactly the incenter and A -excenter of $\triangle AST$ respectively.

Subproof. See that $(A, L; X, Y) \stackrel{D}{=} (K, D; E, F) = -1$ and since $\angle XSY = 90^\circ$, we have $\overline{SX}$ bisects $\angle ASL$ and similarly $\overline{TX}$ bisects $\angle ATL$, which means X is the incenter and also Y is the A -excenter. $\square$

Proof for (a). See that $\overline{AI}$ bisects $\angle AST$ and hence $\overline{AS}$ and $\overline{AT}$ are isogonal lines.

So by $\sqrt{bc}$ inversion at A in $\triangle ABC$ we get $T^* = \overline{AS} \cap (ABC)$ and similarly $S^* = \overline{AT} \cap (ABC)$ and hence (AST) swaps with $\overline{S^*T^*}$ which is parallel to $\overline{BC}$ as required. $\blacksquare$

Proof for (b). We need one more claim.

Claim — (DEF) and $(XTYS)$ are orthogonal.

Subproof. See that it is equivalent to prove that $(G, H; X, Y) = -1$ which is evident after projecting on the incircle by D . $\square$

Again by the Main Claim, we get that midpoint of $\overline{XY}$ actually lies on (AST) by Incenter-Excenter Lemma. And hence inverting about $(XTYS)$ we get

- (DEF) stays put.
- (AST) swaps with $\overline{ST}$.

Obviously both these objects are tangent to each other and we are done. $\blacksquare$

Remark. Did you help him? Regardless, he says “Woof Woof!”, which means “Thank you!” in dog language.

§4.9 Japan TST 2022/6

Problem statement

Given is a circle Γ with diameter MN and a point A inside Γ . The circle with center N , passing through A , meets Γ at B and C . Let $P, Q \in BC$, such that $\angle BAP = \angle QAC$. The lines NP, NQ meet Γ at X, Y , respectively. Prove that AM, PY, QX are concurrent.

Proof. We need to prove some claims before the main proof.



We will apply $\sqrt{bc}$ inversion at A in $\triangle ABC$ (let Z^* be image of Z through this map).

Claim — (APQ) and (ABC) are tangent.

Subproof. See that $\overline{P^*Q^*} \parallel \overline{BC}$ and so (ABC) and (APQ) are tangent at A . $\square$

Noticing that an inversion around (ABC) swaps P with X and Q with Y , and so we get several things.

- (APX) and (AQY) are tangent.
- $\overline{NA}$ is tangent to (AQY) and $\overline{NB}$ is tangent to (BQY) .
- (APQ) , (AXY) and (ABC) are tangent.
- $(XYPQ)$ is cyclic.

Again come back to original picture and see that M is pole of $\overline{BC}$ with respect to (ABC) and so we just need to prove that $\overline{AK^*}$, (AQ^*X^*) , (AP^*Y^*) concur at a second point (where K^* is midpoint of $\overline{BC}$), and so it is sufficient to show that K^* lies on the radical axis of these two circles.

Claim — $\gamma = P^*Q^*Y^*X^*$ is a rectangle with diagonals meeting at K^* .

Subproof. See that from the previous observations, we have that γ is a parallelogram, but since it is cyclic, it means that it is a rectangle.

Now also notice that since $\overline{X^*Y^*} \parallel \overline{BC} \parallel \overline{P^*Q^*}$ and also Y^*X^*BC and P^*Q^*BC are cyclic and hence both are isosceles trapezoids which means that

$$X^*B = Y^*C \text{ and } BP^* = CQ^*$$

Now we will prove that P^*BY^*C is a parallelogram which proves all the lengths above are equal, proving our claim. See that

$$\begin{aligned} \angle Y^*CB &= \angle Y^*CA + \angle ACB = \angle BYA + \angle ACB = \angle BYQ + \angle QYA + \angle ACB \\ &= \angle NBQ + \angle QAN + \angle ACB \\ &= \angle NBC + \angle P^*AN + \angle ACB \\ &= \frac{\angle BNC}{2} + \frac{\angle P^*NA}{2} + \angle ACB \\ &= \angle BAC + \angle ACP^* + \angle ACB \\ &= \angle CBA + \angle ACP^* \\ &= \angle CP^*A + \angle ACP^* \\ &= \angle P^*AC = \angle P^*BC \end{aligned}$$

which proves that $\overline{Y^*C} \parallel \overline{P^*B}$ and we can similarly painfully show $\overline{Y^*B} \parallel \overline{P^*C}$. $\square$

But see that from the claims we proved, we get that and hence we get that

$$K^*X^* \cdot K^*Q^* = K^*P^* \cdot K^*Y^*$$

and we win. $\blacksquare$

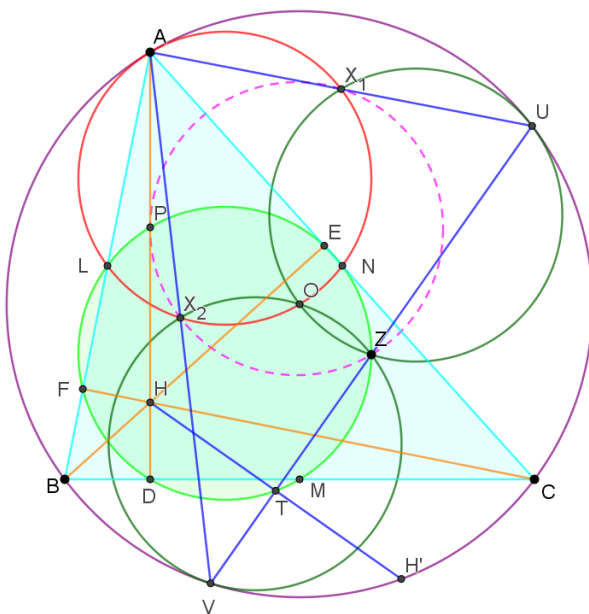
§4.10 Taiwan TST Round 3 2014/3

Problem statement

Let ABC be a triangle with circumcircle Γ and let M be an arbitrary point on Γ . Suppose the tangents from M to the incircle of ABC intersect $\overline{BC}$ at two distinct points X_1 and X_2 . Prove that the circumcircle of triangle MX_1X_2 passes through the tangency point of the A -mixtilinear incircle with Γ .

Proof. We will invert about the incircle and using the fact that the mixtilinear incircle, incenter and major arc midpoint of $\widehat{BC}$ are collinear, we can exploit the incenter-orthocenter duality to get this new problem with some reformed labels.

Let $\triangle ABC$ be a triangle with O circumcenter, H orthocenter, Γ circumcircle, Ω nine point circle, $\triangle DEF$ orthic triangle and $\triangle MNL$ medial triangle. Let $Z \in \Omega$ be an arbitrary point and U and V be the unique distinct points such that (OZU) and (OZV) are both tangent to Γ . Let $X_1 = (ALON) \cap (OZU)$ and define X_2 similarly. Prove that if P is the midpoint of $\overline{AH}$, then X_1, Z, X_2, P are concyclic.



See that because of Pole Polar/La Hire shenanigans, we have that $\triangle X_1ZX_2$ is the medial triangle of $\triangle AUV$.

Hence we have that $\angle X_2ZX_1 = \angle UAV$ and $\angle X_2PX_1 = \angle VHU$ (by taking a double homothety at A) and so if we prove that the reflection of H over $\overline{UV}$ lies on Γ or equivalently proving if T is the unique point on Γ such that $\angle OZT = 90^\circ$ and then $\overline{HT} \parallel \overline{OZ}$, then we are done.

But this is trivial if we take a double homothety at H . ■

§4.11 IGO 2023/A1**Problem statement**

We are given an acute triangle ABC . The angle bisector of $\angle BAC$ cuts BC at P . Points D and E lie on segments AB and AC , respectively, so that $BC \parallel DE$. Points K and L lie on segments PD and PE , respectively, so that points A, D, E, K, L are concyclic. Prove that points B, C, K, L are also concyclic.

Proof. We are obviously $\sqrt{bc}$ invert at A in $\triangle ABC$ like a good boy. See these stuff will happen now.

- $\overline{D^*E^*} \parallel \overline{BC}$.
- P swaps with the minor arc midpoint of $\widehat{BC}$, call it M .
- $K^* = \overline{D^*E^*} \cap (AMD^*)$ and $L^* = \overline{D^*E^*} \cap (AME^*)$.

Claim — M lies on the perpendicular bisector of both $\overline{BC}$ and $\overline{K^*L^*}$.

Subproof. The first is obvious. The next one is just a simple angle chase

$$\angle MK^*D^* = \angle MAD^* = \angle E^*AM = \angle E^*L^*M$$

And done. □

This basically proves BCK^*L^* is an isosceles trapezoid and so it is cyclic. ■

§4.12 4-G-UnS-2**Problem statement**

Let $\triangle ABC$ be a triangle and define Ω_a to be the Apollonian circle passing through A and foci B and C . Define Ω_b and Ω_c similarly. Prove that $\Omega_a, \Omega_b, \Omega_c$ intersect at 2 points. The “inside” intersection is the *First Isodynamic Point* or X_{15} . So further prove that X_{15} is the isogonal conjugate to the *First Fermat Point* or X_{13} .

Proof. We will define some new objects.

- Let I be the incenter of $\triangle ABC$.
- Let $D = \overline{AI} \cap \overline{BC}$ and define E and F similarly.
- Let X be the point on $\overline{BC}$ such that $(B, C; D, X) = -1$. Define Y and Z similarly and hence ω_a is the circle with diameter $\overline{DX}$ and so on.
- Let M_A and N_A be the minor and major arc midpoints of $\widehat{BC}$. Define M_B, M_C, N_B, N_C similarly.

Finally we will $\sqrt{bc}$ invert at A in $\triangle ABC$ and hence we get that

- Ω_a swaps with $\overline{M_A N_A}$.
- Ω_b swaps with the circle with center B and radius $\overline{BC}$. This is because the diameter of the circle must be on the isogonal line of $\overline{AC}$ which is $\overline{AB}$ and see that $(A, C; E, Y) = (\infty, B; E^*, Y^*) = -1$.

But see that the radius of Ω_b^* and Ω_c^* are both $\overline{BC}$ and hence their radical axis is the perpendicular bisector of $\overline{BC}$ which is exactly $\overline{M_A N_A}$.

Now if S is the unique point outside the triangle such that $\triangle SBC$ is equilateral then it is well known that A, S, X_{13} are collinear. But see that S is the “outside” intersection point of the circle at B and C with radii $\overline{BC}$ which is exactly X_{15}^* and hence $\overline{AX_{15}}$ and $\overline{AX_{13}}$ are isogonal and do same for other vertices and second part is also proven. ■

§4.13 4-G-UnS-3**Problem statement**

Let there be $\triangle ABC$ with circumcenter O and incenter I . Let ω be the circle externally tangent to (BOC) and tangent to $\overline{AB}$ and $\overline{AC}$ at say M and N respectively. Prove that $\overline{MN}$ bisects $\overline{AI}$.

Proof. Dilate everything by a factor of 2 around A and call Ω the image of ω and Γ the image of (BOC) .

The main claim is this after which we will be done.

Claim — Ω is the A -mixtilinear incircle.

Subproof. Due to phantom object shenanigans, we just need to prove that the A -mixtilinear incircle is tangent to Γ .

We will now $\sqrt{bc}$ invert at A in $\triangle ABC$, and see that the A -mixtilinear incircle maps to the A -excircle and Γ actually maps to the 9-point circle of $\triangle ABC$, and these two circles are tangent by the result of Feuerbach! $\square$

We are obviously done now because it is well known that the touch chord of the mixtilinear incircle passes through I , as desired. $\blacksquare$